

TASK3 DOCUMENTATION

Project Title:

SQL Data Analysis Using SQLite3 on Dataset for Data Analytics

Abstract

Structured Query Language (SQL) is widely used for managing and analyzing data stored in databases. This project focuses on performing SQL Data Analysis on the "Dataset for Data Analytics" using SQLite3 in Jupyter Notebook. The cleaned dataset was imported into a SQLite database, and various SQL queries were executed to retrieve, filter, group, and summarize data. The project demonstrates how SQL can be used to transform raw data into meaningful information and support data-driven decision-making.

1.Introduction

Databases are essential for storing and managing large volumes of data. SQL provides an efficient way to retrieve, manipulate, and analyze data stored in databases. Through SQL queries, users can extract specific information, calculate metrics, identify trends, and generate reports.

This project focuses on performing SQL Data Analysis using SQLite3 within Jupyter Notebook. The cleaned dataset was imported into a SQLite database, and various SQL operations were performed to analyze the data. The project highlights the practical application of SQL in data analytics and business reporting.

2. Objectives

- To create a SQL database for storing the dataset.
- To import the cleaned dataset into a database table.
- To retrieve data using SQL queries.
- To filter records using SQL conditions.
- To group records using SQL aggregation techniques.
- To calculate summary statistics using aggregate functions.
- To derive meaningful insights from the dataset.

3. Problem Statement

Organizations generate and store large amounts of data daily. Analyzing this data manually is time-consuming and inefficient. SQL provides a powerful mechanism for querying and summarizing data efficiently. The objective of this project is to use SQLite3 and SQL queries to retrieve, organize, and analyze data stored in a database and generate valuable insights for decision-making.

4. Dataset Description

Dataset Name: cleaned_dataset

The dataset used in this project was originally obtained from a previous dataset. It was cleaned and preprocessed to improve data quality, and the cleaned data was then exported to a new Excel file. This exported dataset was imported into a SQLite database and analyzed using SQL queries to generate meaningful insights.

Dataset Information

- Dataset Name: cleaned_dataset
- File Format: Excel (.xlsx)
- Database Used: SQLite3
- Domain: Data Analytics
- Purpose: SQL Data Analysis

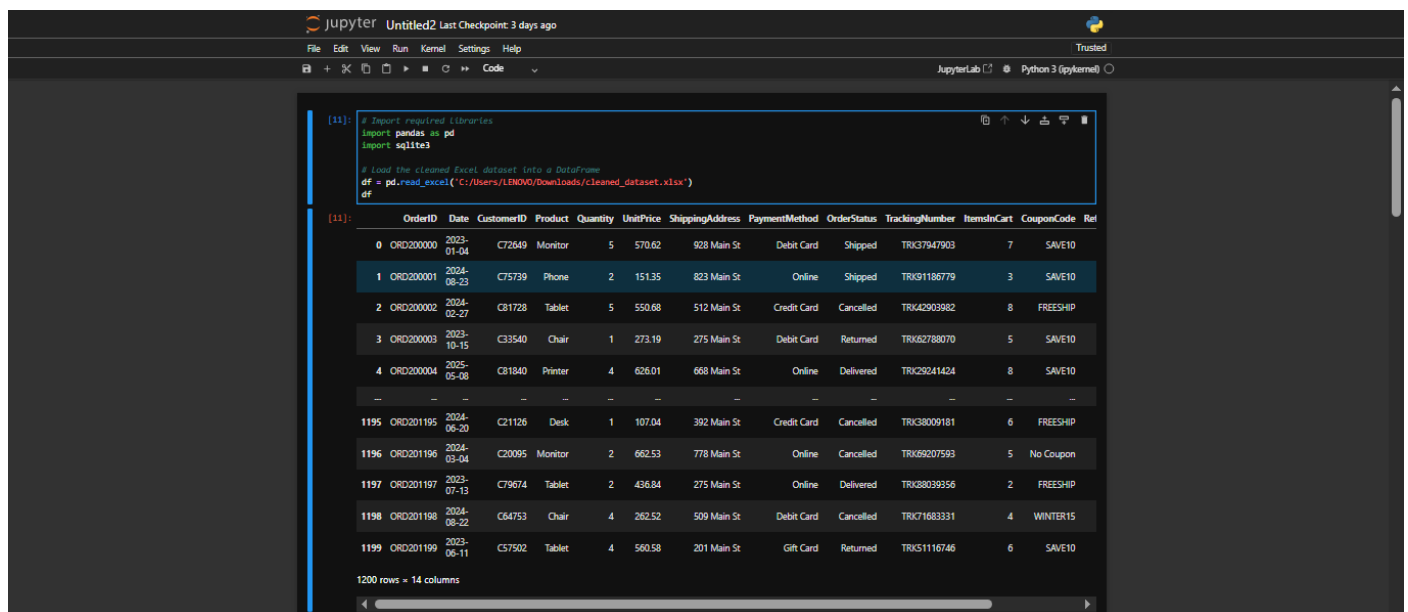
5. Tools and Technologies Used

- SQLite3
- SQL
- Python
- Pandas
- Jupyter Notebook
- Microsoft Excel

6. Methodology

Step 1: Dataset Loading

The cleaned dataset was loaded into Jupyter Notebook using Pandas.



```
(11): # Import required libraries
import pandas as pd
import sqlite3

# Load the cleaned Excel dataset into a DataFrame
df = pd.read_excel("C:/Users/LENOVO/Downloads/cleaned_dataset.xlsx")
df
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	Rel
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRX91186779	3	SAVE10	
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	
...	...	...	...	...	...	...	...	...	...	...	...	...	...
1195	ORD201195	2024-06-20	C21126	Desk	1	107.04	392 Main St	Credit Card	Cancelled	TRK38009181	6	FREESHIP	
1196	ORD201196	2024-03-04	C20095	Monitor	2	662.53	778 Main St	Online	Cancelled	TRK69207593	5	No Coupon	
1197	ORD201197	2023-07-13	C79674	Tablet	2	436.84	275 Main St	Online	Delivered	TRK88039356	2	FREESHIP	
1198	ORD201198	2024-08-22	C64753	Chair	4	262.52	509 Main St	Debit Card	Cancelled	TRK71683331	4	WINTER15	
1199	ORD201199	2023-06-11	C57502	Tablet	4	560.58	201 Main St	Gift Card	Returned	TRK51116746	6	SAVE10	

1200 rows x 14 columns

Step 2: Database Creation

A SQLite database was created using the SQLite3 library.

Step 3: Data Import

The dataset was imported into a SQLite table for SQL analysis.

```
[12]: # Create a connection to the SQLite database
conn = sqlite3.connect('sales.db')

# Import the DataFrame into the SQLite database
df.to_sql(
    name='sales',
    con=conn,
    if_exists='replace',
    index=False
)

print("Dataset imported successfully!")
```

Dataset imported successfully!

Step 4: Query Execution

SQL queries were executed to retrieve information from the database.

```
[13]: # Define an SQL query to retrieve the first 5 records

query = "SELECT * FROM sales LIMIT 5"

result = pd.read_sql(query, conn)

print(result)
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	\
0	ORD200000	2023-01-04 00:00:00	C72649	Monitor	5	570.62	
1	ORD200001	2024-08-23 00:00:00	C75739	Phone	2	151.35	
2	ORD200002	2024-02-27 00:00:00	C81728	Tablet	5	550.68	
3	ORD200003	2023-10-15 00:00:00	C33540	Chair	1	273.19	
4	ORD200004	2025-05-08 00:00:00	C81840	Printer	4	626.01	

	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	\
0	928 Main St	Debit Card	Shipped	TRK37947903	7	
1	823 Main St	Online	Shipped	TRK91186779	3	
2	512 Main St	Credit Card	Cancelled	TRK42903982	8	
3	275 Main St	Debit Card	Returned	TRK62788070	5	
4	668 Main St	Online	Delivered	TRK29241424	8	

	CouponCode	ReferralSource	TotalPrice
0	SAVE10	Instagram	2853.10
1	SAVE10	Referral	302.70
2	FREESHIP	Email	2753.40
3	SAVE10	Facebook	273.19
4	SAVE10	Email	2504.04

Step 5: Data Filtering

WHERE clauses were used to filter records based on specific conditions.

```
[6]: # Data Filtering using WHERE Clause

query = """
SELECT OrderID,Product,TotalPrice
FROM sales
WHERE TotalPrice > 2500
"""

print(pd.read_sql(query, conn))
```

	OrderID	Product	TotalPrice
0	ORD200000	Monitor	2853.10
1	ORD200002	Tablet	2753.40
2	ORD200004	Printer	2504.04
3	ORD200010	Tablet	3129.85
4	ORD200032	Tablet	2683.60
..	...	...	...
89	ORD201142	Phone	2780.40
90	ORD201144	Desk	2539.70
91	ORD201153	Monitor	2786.84
92	ORD201156	Laptop	2763.12
93	ORD201179	Laptop	2592.75

[94 rows x 3 columns]

Step 6: Data Grouping

GROUP BY clauses were used to categorize records and generate summarized results.

```
[15]: # Calculate total quantity sold for each product
```

```
query = """
SELECT Product,
       SUM(Quantity) AS Total_Quantity
FROM sales
GROUP BY Product
"""

print(pd.read_sql(query, conn))
```

	Product	Total_Quantity
0	Chair	562
1	Desk	508
2	Laptop	535
3	Monitor	480
4	Phone	411
5	Printer	542
6	Tablet	497

Step 7: Aggregation Analysis

Aggregate functions were applied to calculate important metrics, including:

- COUNT()
- SUM()
- AVG()

```
[7]: # Count total number of records in the sales table
```

```
query = """
SELECT COUNT(*) AS Total_Orders
FROM sales
"""

print(pd.read_sql(query, conn))
```

	Total_Orders
0	1280

```
•[4]: # Calculate total sales amount
```

```
query = """
SELECT SUM(TotalPrice) AS Total_Revenue
FROM sales
"""

print(pd.read_sql(query, conn))
```

	Total_Revenue
0	1264761.96

```
•[5]: # Calculate average sales amount
```

```
query = """
SELECT AVG(TotalPrice) AS Average_Order_Value
FROM sales
"""

print(pd.read_sql(query, conn))
```

	Average_Order_Value
0	1053.9683

Step 8: Data Sorting

ORDER BY clauses were used to sort query results in ascending or descending order.

```
[16]: # Calculate total orders for each payment method
      # and display payment methods based on order frequency
```

```
query = """
SELECT PaymentMethod,
       COUNT(*) AS Total_Orders
FROM sales
GROUP BY PaymentMethod
ORDER BY Total_Orders
"""

print(pd.read_sql(query, conn))
```

	PaymentMethod	Total_Orders
0	Gift Card	230
1	Debit Card	232
2	Credit Card	234
3	Cash	246
4	Online	258

Step 9: Interpretation

The results obtained from SQL queries were analyzed and interpreted to derive meaningful insights.

7. SQL Operations Performed

Data Retrieval

- Retrieved all records using SELECT statements.
- Retrieved specific columns for analysis.

Data Filtering

- Applied WHERE conditions to extract relevant records.
- Filtered data based on user-defined criteria.

Data Grouping

- Grouped records using GROUP BY.
- Generated category-wise summaries.

Aggregate Functions

- Used COUNT() to determine the number of records.
- Used SUM() to calculate total values.
- Used AVG() to calculate average values.

Data Sorting

- Sorted results using ORDER BY.
- Displayed records in ascending and descending order.

8. Results

The SQL Data Analysis process successfully generated meaningful information from the dataset.

The analysis resulted in:

- Efficient retrieval of records from the database.
- Accurate filtering of relevant data.
- Group-wise summaries using aggregation functions.
- Statistical calculations using SQL functions.
- Improved understanding of the dataset through structured queries.

9. Key Observations

- SQL simplifies the process of data retrieval and analysis.
- SQLite3 provides an easy-to-use database environment within Jupyter Notebook.
- Aggregate functions help summarize large datasets efficiently.
- Grouping operations provide meaningful category-wise insights.
- Filtering operations enable focused analysis on specific subsets of data.

10. Key Skills Gained

- SQLite Database Management
- SQL Query Writing

- Data Import and Export
- Data Filtering using WHERE Clause
- Data Grouping using GROUP BY Clause
- Aggregate Functions (COUNT, SUM, AVG)
- Result Sorting using ORDER BY Clause
- Data Analysis and Interpretation

11. Challenges Faced

- Understanding SQL syntax and query structure.
- Importing data into the SQLite database.
- Writing accurate filtering conditions.
- Interpreting aggregated results.
- Managing large query outputs effectively.

12. Conclusion

The SQL Data Analysis project successfully demonstrated how SQLite3 and SQL can be used to analyze structured data efficiently. Various SQL operations such as SELECT, WHERE, GROUP BY, ORDER BY, COUNT(), SUM(), and AVG() were applied to generate meaningful insights from the dataset. The project enhanced understanding of database management and SQL-based data analysis techniques.

13. References

1. SQLite3 Documentation
2. SQL Documentation
3. Pandas Documentation
4. Jupyter Notebook Documentation
5. DecodeLabs Industrial Training Kit
6. Data Analytics Course Materials